

Human-centred AI as a Design Process

Part 1: Understanding user needs
& Defining the right problem

Dr Georgia Panagiotidou

Overview

What is Human-Centred Design?

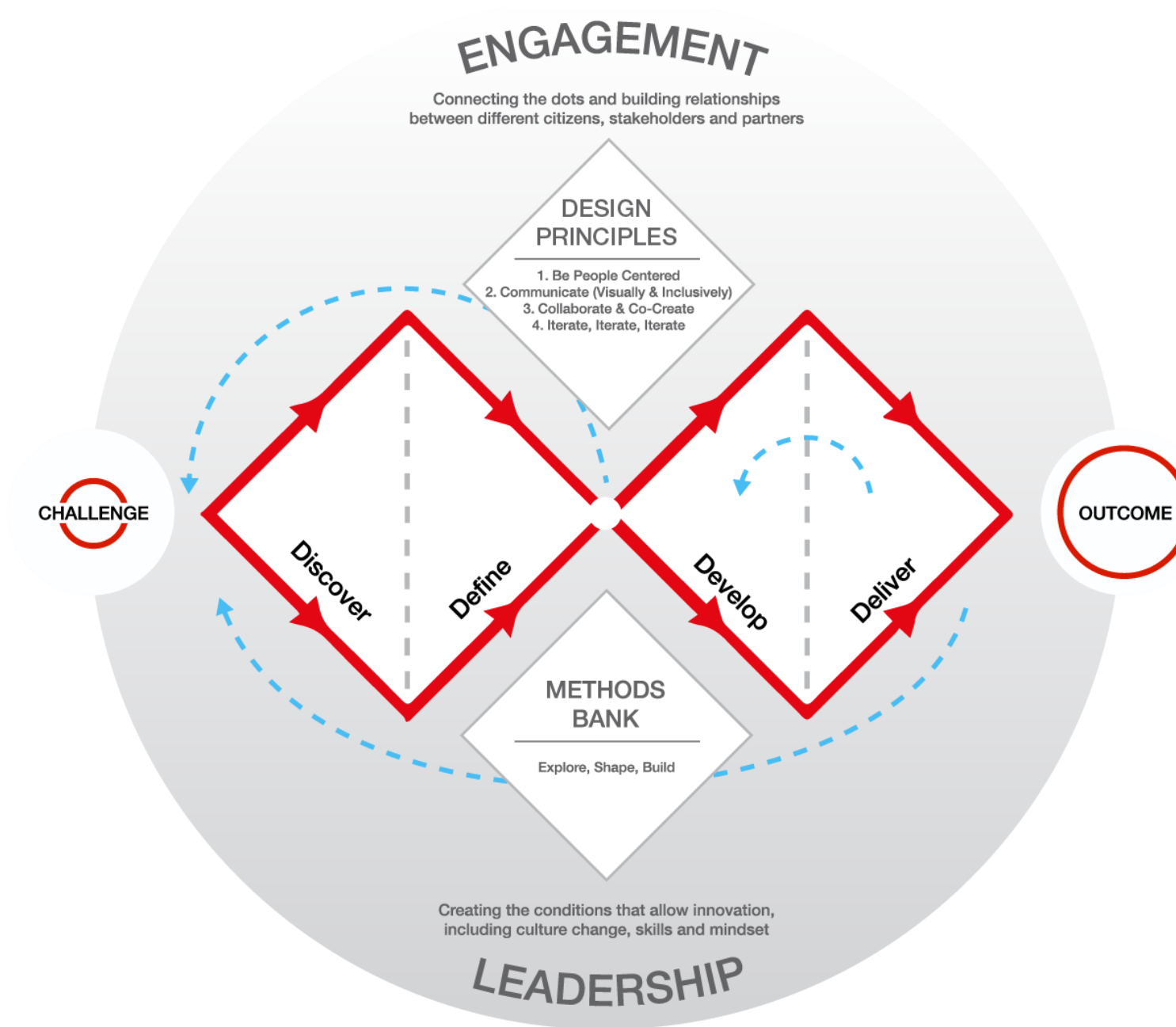
- Who to involve in the design process and why
- Degrees and moments of user involvement
- Four basic activities of the design process
- A simple lifecycle model for the design process

The process supports understanding:

- How to find out what people need
- How to decide what to design
- How to generate alternative designs
- How to choose among alternative designs

What is Human-Centred Design?

- It is a process that is:
 - Focused on discovering requirements, designing to fulfil requirements, producing prototypes and evaluating them
 - Traditionally focused on users and their goals although with HAI this is somewhat evolving
 - Involves trade-offs to balance conflicting needs and requirements



INSPIRATION

I have a design challenge.

How do I get started?
How do I conduct an interview?
How do I stay human-centered?

IDEATION

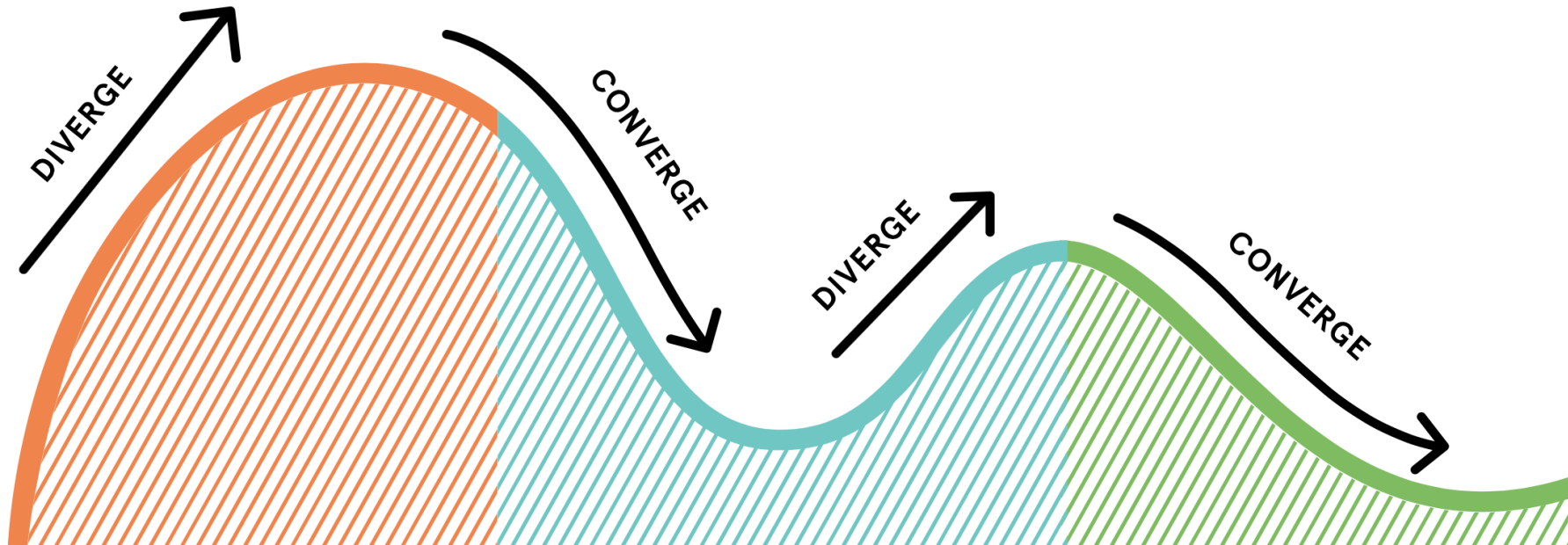
I have an opportunity for design.

How do I interpret what I've learned?
How do I turn my insights into tangible ideas?
How do I make a prototype?

IMPLEMENTATION

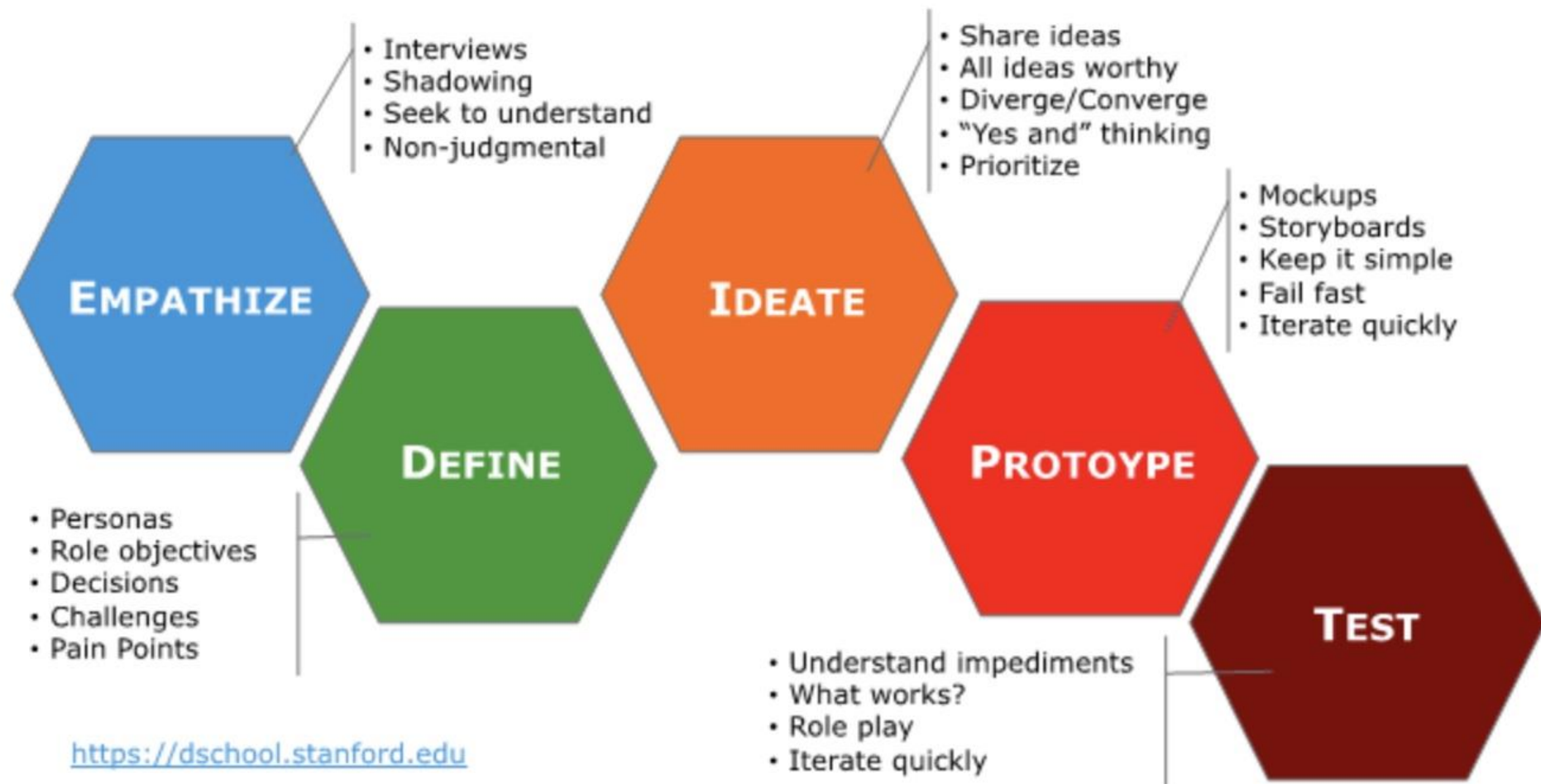
I have an innovative solution.

How do I make my concept real?
How do I assess if it's working?
How do I plan for sustainability?



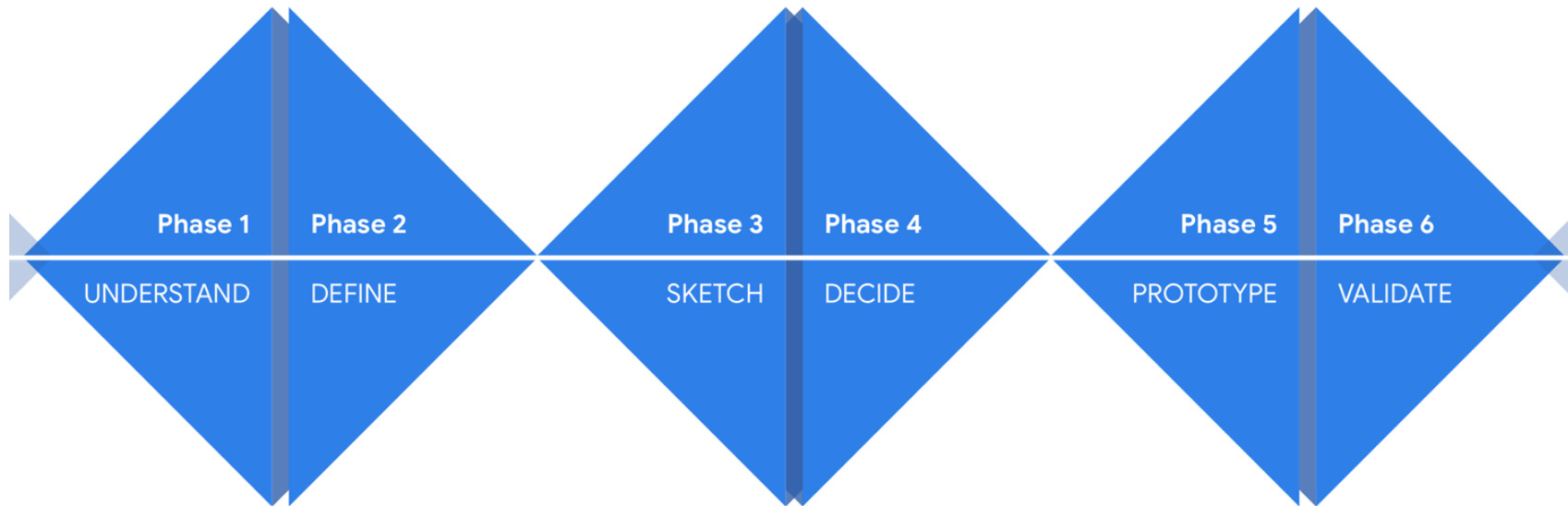
IDEO

Stanford d.school Design Thinking Process



Other lifecycle models:

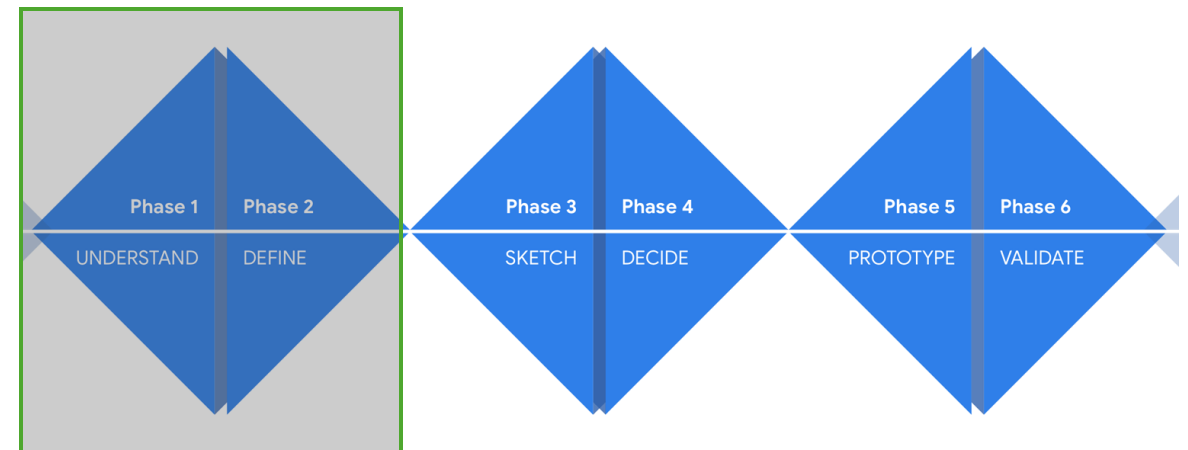
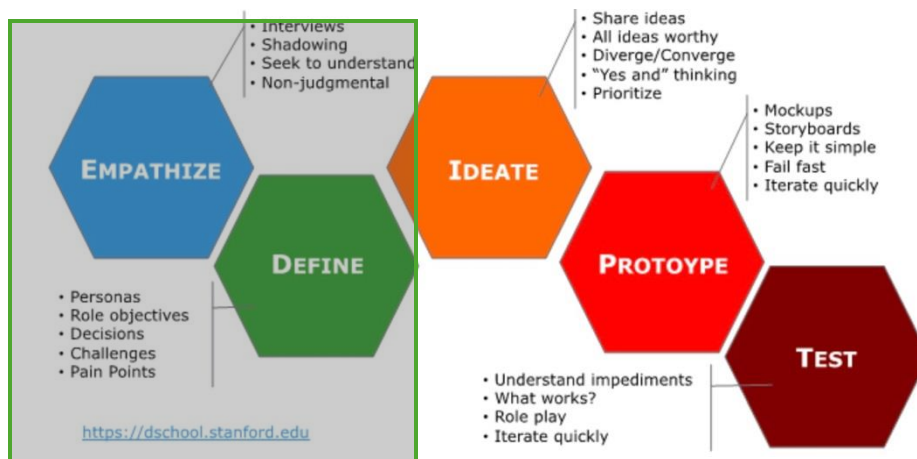
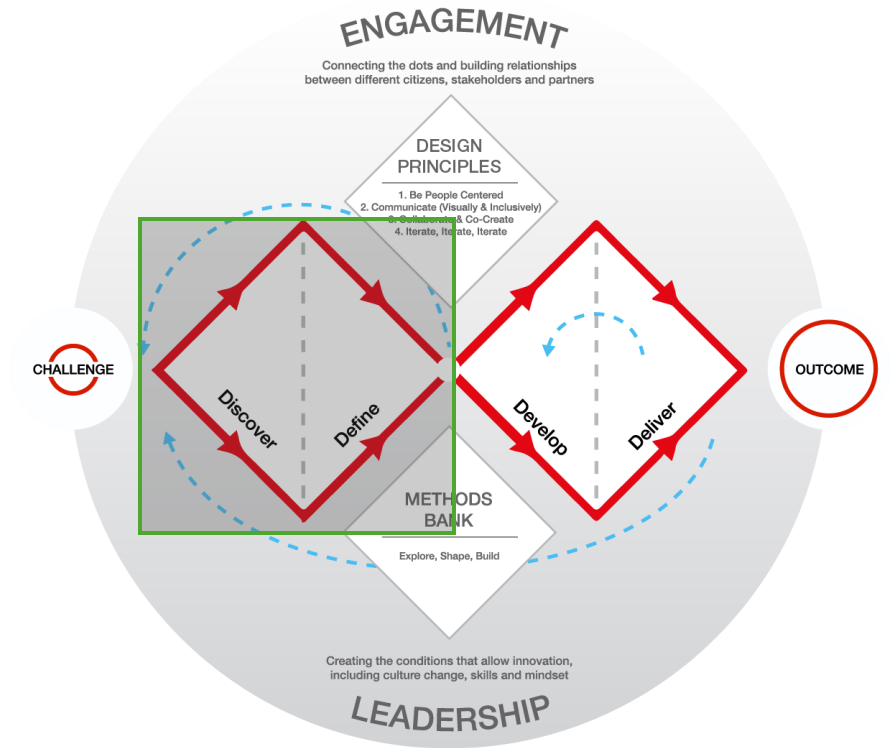
Design Sprints



designsprintkit.withgoogle.com/methodology/overview

Today's focus

- Discover / Empathise / Understand
- Define



Importance of involving stakeholders

Expectation management

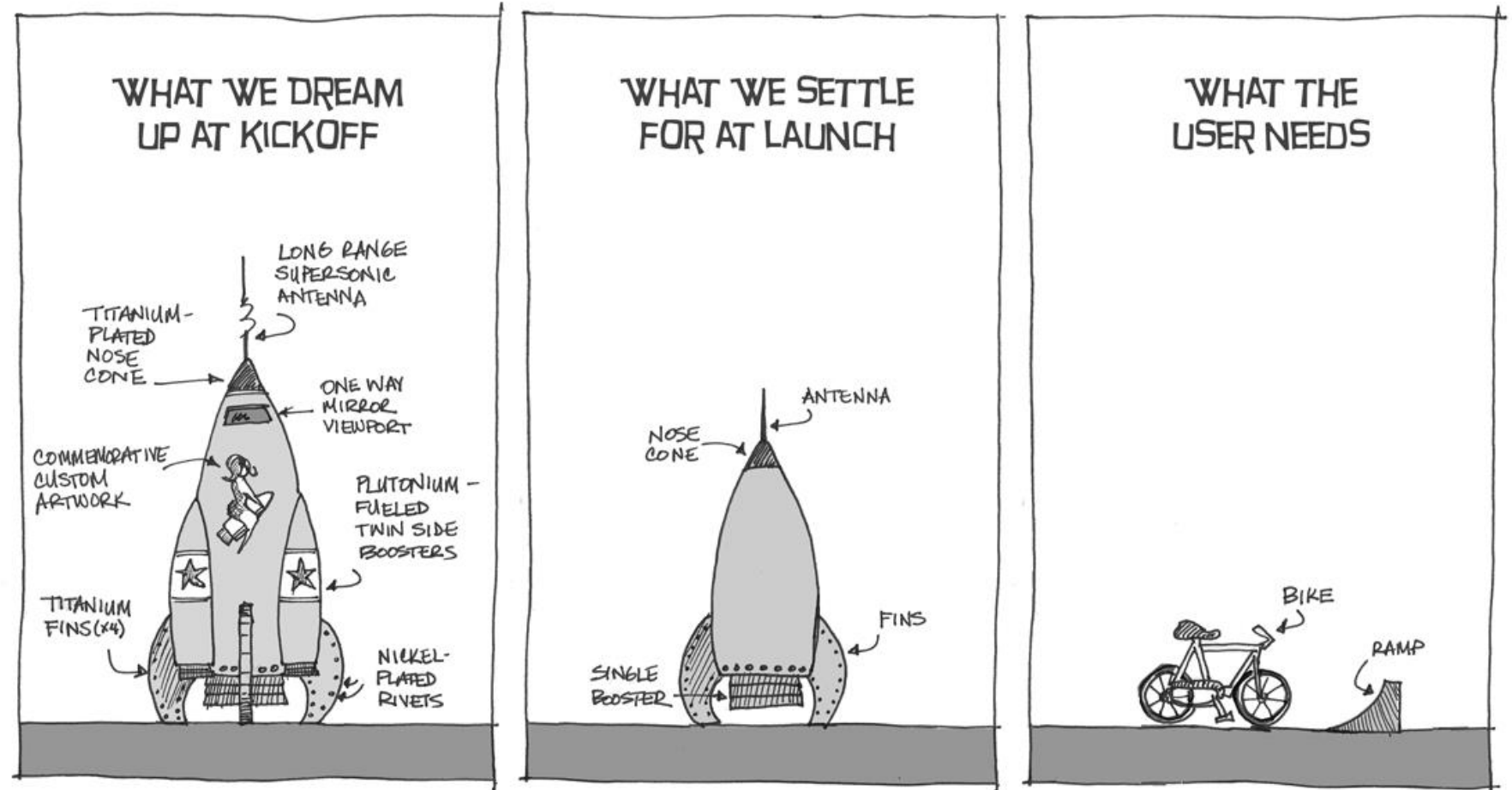
- Realistic expectations
- No surprises, no disappointments
- Timely training
- Communication, but no hype

Ownership

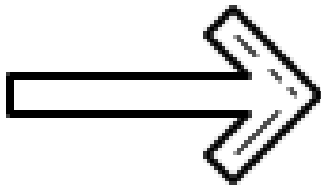
- Make the users active stakeholders
- More likely to forgive or accept problems
- Can make a big difference in acceptance and success of product

Discover: understanding users' needs

THE UX DESIGNER PARADOX



... so you don't
end up
with this



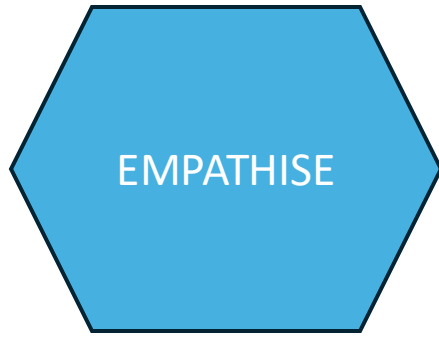
DISCOVER
User needs

BONUS 2015

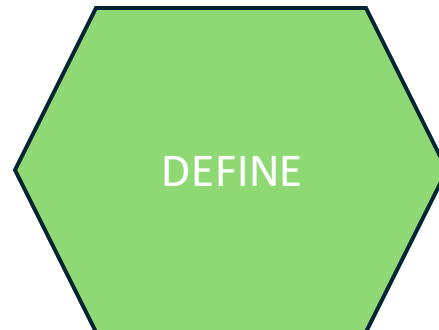
Degrees of stakeholder involvement

- Member of the design team
- Small group or individual activities
- Online contributions from thousands of users
 - Online Feedback Exchange (OFE) systems
 - Crowdsourcing design ideas
 - Citizen engagement
- Participatory design
- User involvement after product release
 - A/B testing
 - Customer reviews

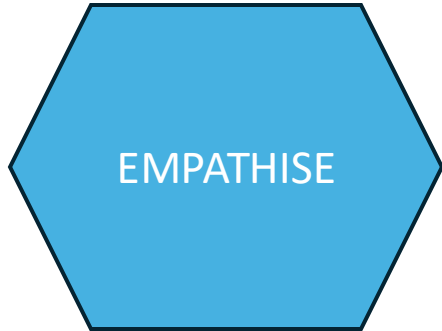
How do needs and requirements interact?



1. To understand as much as possible about the **users**, their **activities**, and the **context** of that activity, so the system under development can support them in achieving their goals

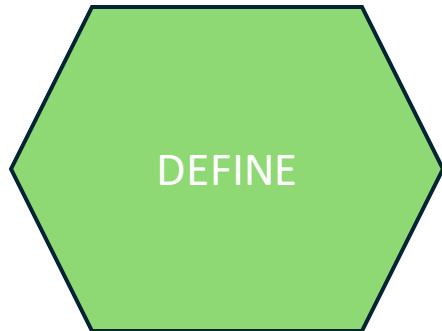


2. To produce a **set of initial requirements** that form a sound basis to start designing



1. To understand as much as possible about the **users**, their **activities**, and the **context** of that activity, so the system under development can support them in achieving their goals

what are requirements?



A requirement is a statement about an intended product that specifies what it should do or how it should perform. One of the aims of the requirements activity is to make the requirements as specific, unambiguous, and clear as possible.

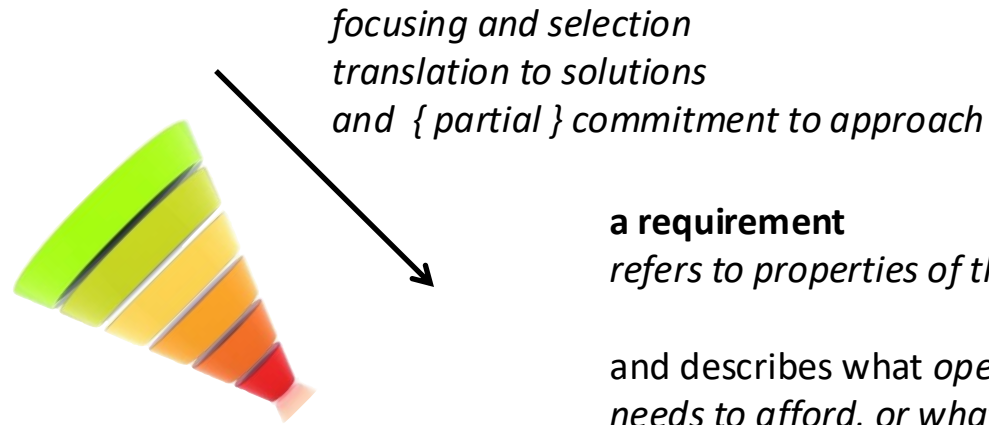
(from Interaction Design: Beyond Human-Computer Interaction 4th ed, p353)

a need

takes a *user's perspective* (of the problem to be solved)

and describes *constraints, goals, hopes,*
and activities performed by the user

is often contextual, embedded,
and social



a requirement

refers to properties of the system (to solve the problem)

and describes what *operations the system*
needs to afford, or what characteristics
and properties it needs to have, and
how it should operate

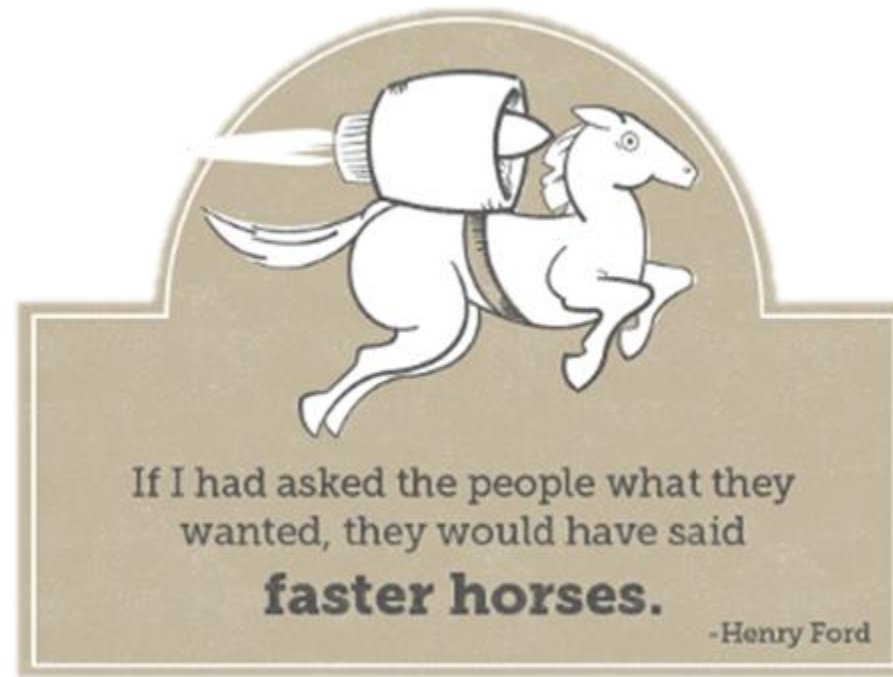
usually embodies or at least partially commits to one
particular solution, philosophy or
approach to addressing needs

there may be many strategies approaches to address a
user's need

How do we { elicit, discover, understand } needs?



How do we { elicit, discover, understand } users' needs?



Ways of identifying stakeholder needs:

Observation (direct / indirect)

Interviews

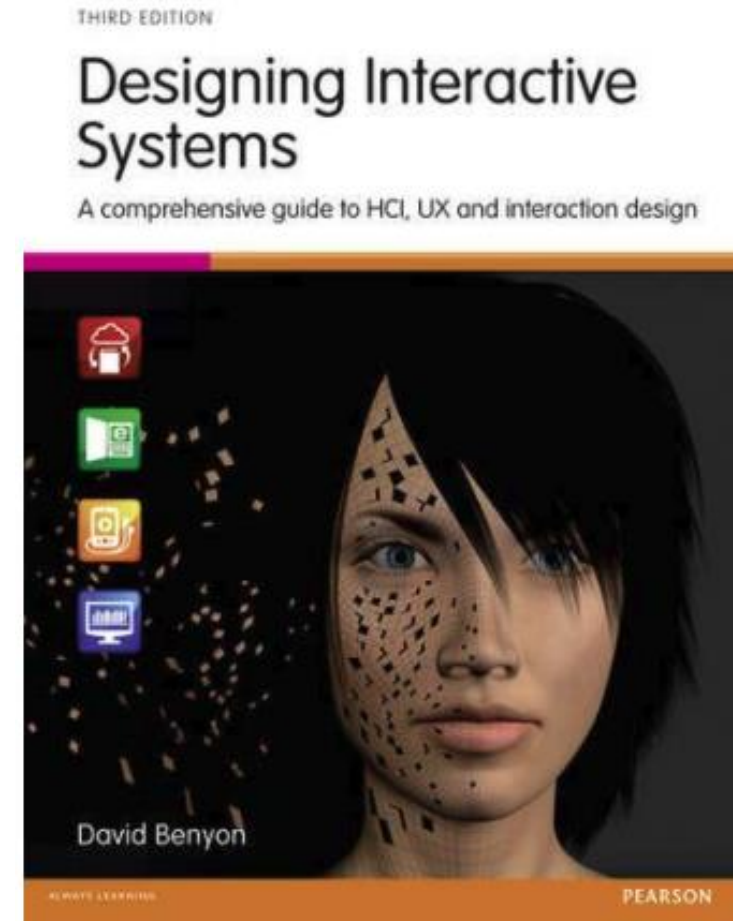
Questionnaires

(Digital) ethnography

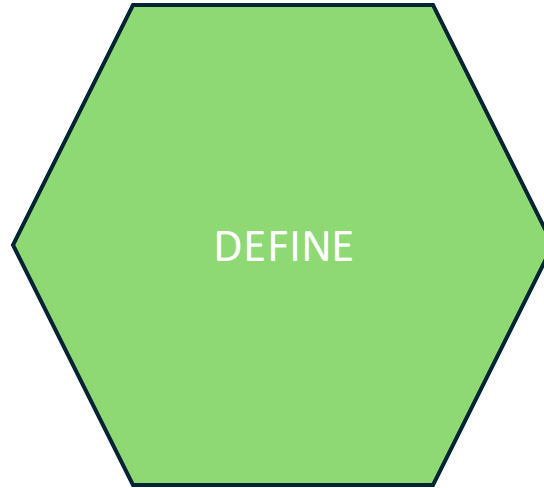
Technology probes

Participant Observation (Field studies as a Participant Observer)

Critical Design



Chapter 7



Moving from **individual observations** to **generalised insights**

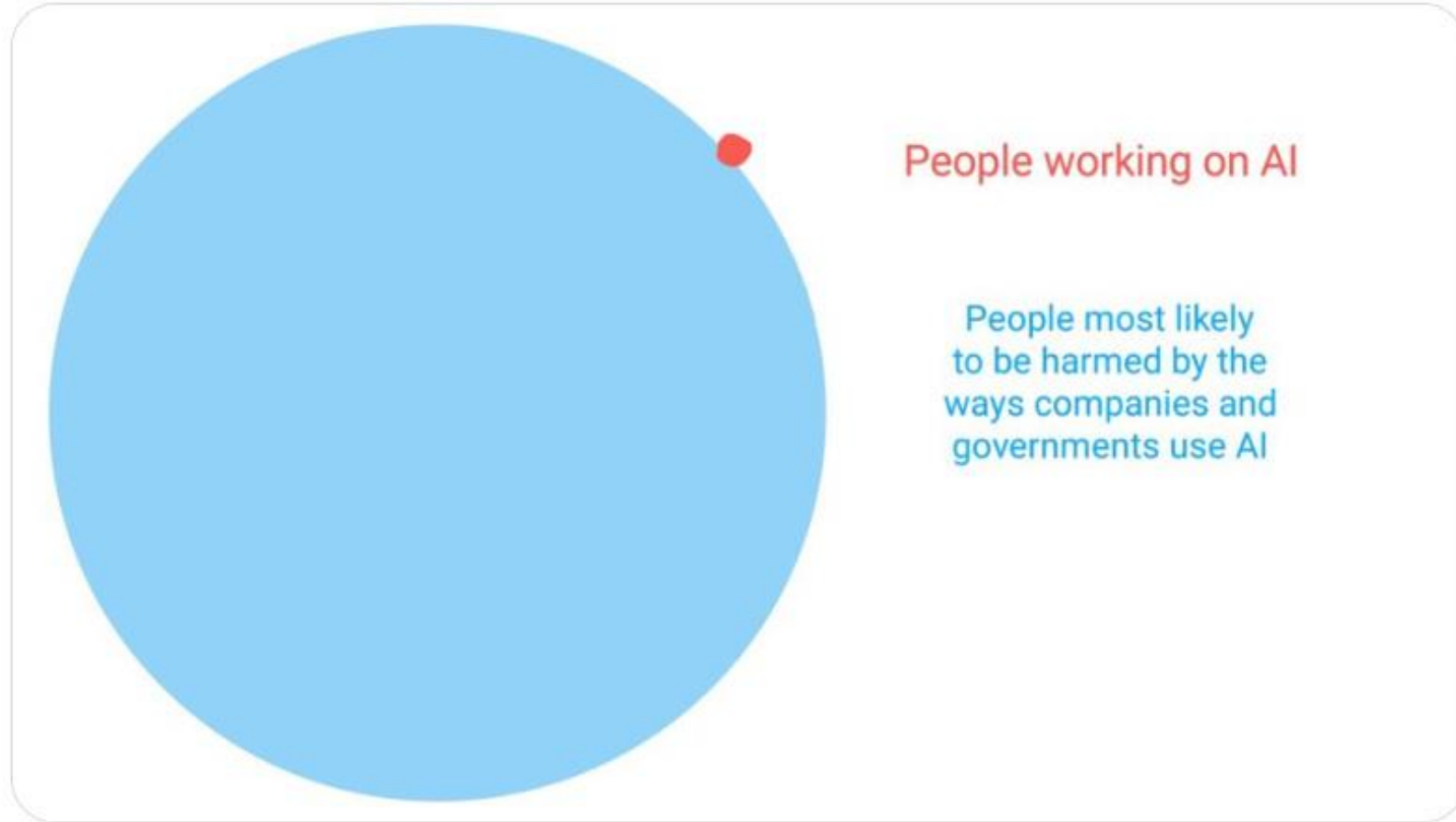
... that help us **focus** on the '**right**' design problem

What does this child need?

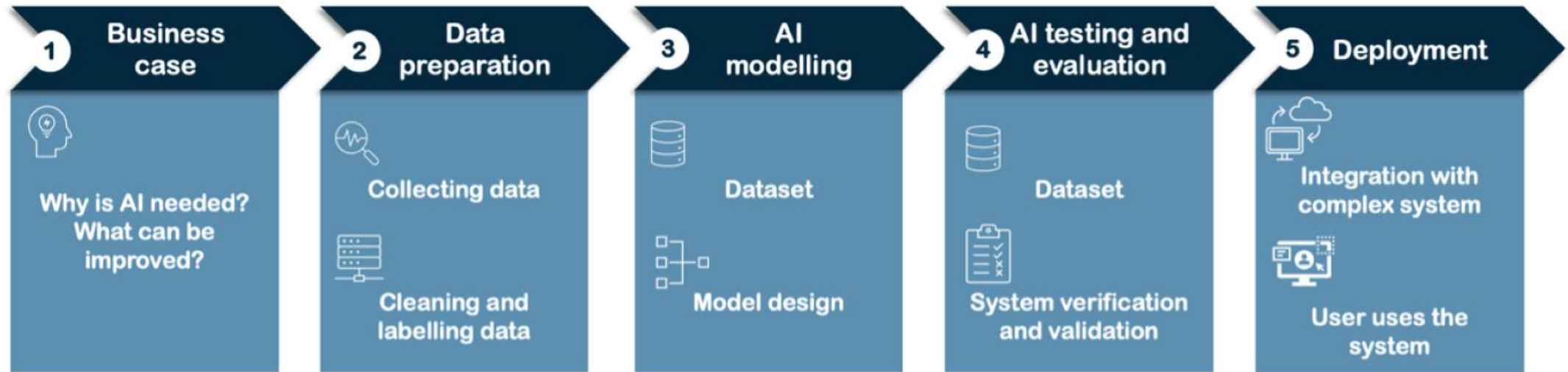




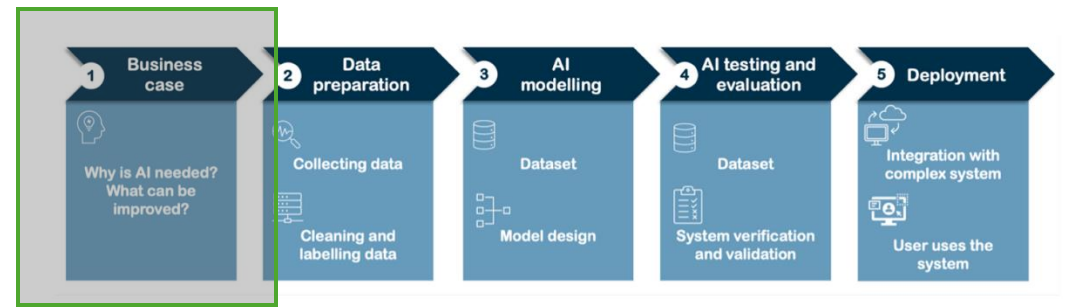
Sasha Costanza-Chock @schock · 5 Nis 2023



Stakeholder involvement in developing Human-AI systems

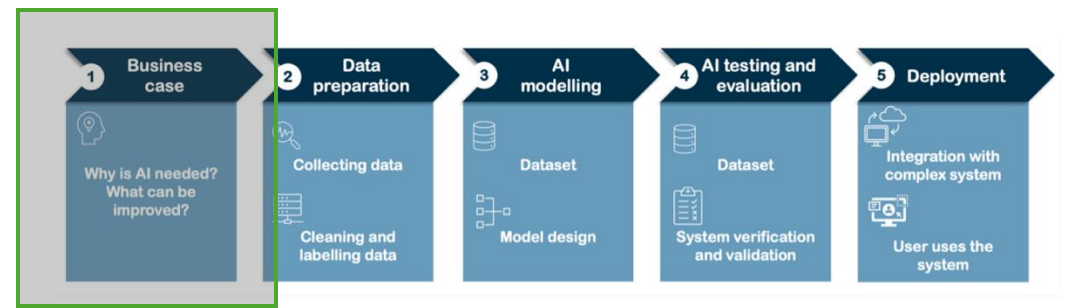


Early in the process



- Consider if the problem you're solving requires an AI solution
- Decide if AI adds unique value
- Balance a **people-first approach** and a **technology-first approach**

Early in the process



When AI is probably better

- Recommending different content to different users
- Prediction of future events
- Personalization improves the user experience
- Natural language understanding.
- Recognition of an entire class of entities.
- Detection of low occurrence events that change over time.
- An agent or bot experience for a particular domain.
- Showing dynamic content is more efficient than a predictable interface.

When AI is probably not better

- Maintaining predictability
- Providing static or limited information
- Minimizing costly errors.
- Complete transparency
- Optimizing for high speed and low cost.
- Automating high-value tasks.

Observation (direct / indirect)

- How was this work done before AI?

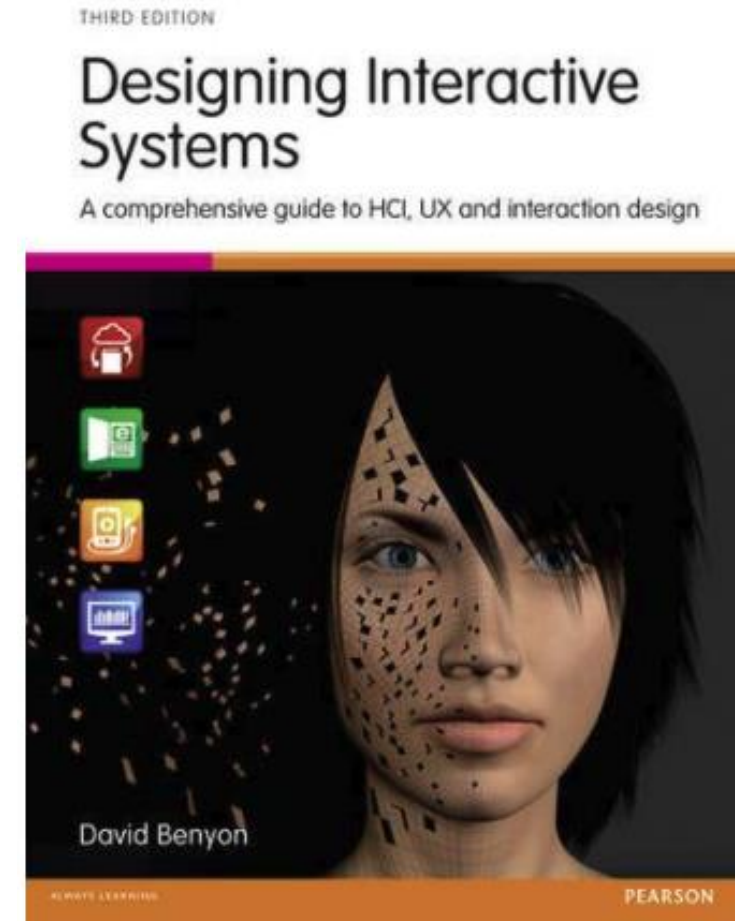
Interviews

- What are people's mental models about the models/processes under consideration?

Questionnaires

(Digital) ethnography

- What perceptions do people have of similar products?
- How was this work done before AI?



Chapter 7

1. Statement of User Need

Write down a clear, focused statement of the user need:

How might we solve _____ { our user need } _____?

How can AI solve this problem in a unique way?

When considering if and how AI might solve the problem in a unique way, consider:

How pervasive is the problem?	
Whom does it affect?	
How does this problem affect various individuals?	
How does this problem vary across different dimensions of identity or historically marginalized communities?	
Do the problem's effects change over time?	

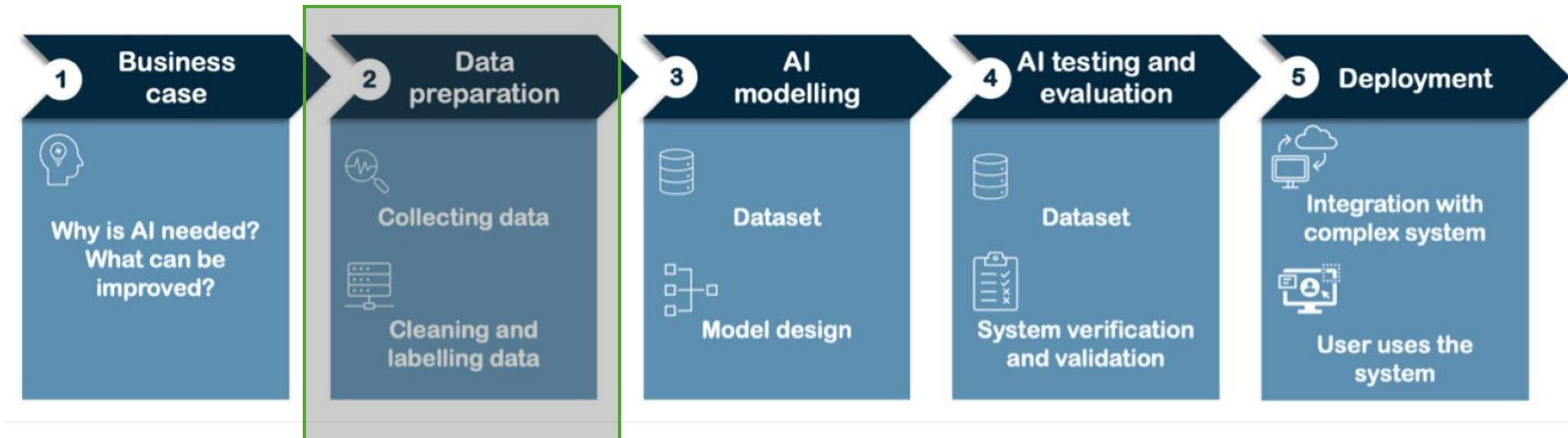
2. AI suitability

Read through each of our “AI suitability” statements to identify if your user need is a potential good fit for an AI solution. You may add one or two more statements that describe your team’s specific AI capabilities.

AI is probably better for	AI is probably not better for
☐ Personalization: The core experience requires personalizing content to individual users, or to a domain	☐ Consistency: The most valuable part of the core experience is its predictability, regardless of context or input
☐ Predictions: The core experience requires prediction of future events	☐ Cost: Running an AI solution at the needed frequency is very high
☐ Scale: Need to recognize a general class of things that is too large to articulate every case	☐ Stakes: The consequences of errors is very high and outweighs the benefits of a small increase in success rate
☐ Volume: Need to detect low occurrence events that are constantly evolving	☐ Explainability: Users, customers, or developers need to understand exactly everything that happens in the code and application
☐ Language: User experience requires natural language interactions, or agent or bot experience for a particular domain	☐ Momentum: Responsibly getting to market first is more important than other factors or the value using AI would provide
☐ Diversification: The user experience is significantly enhanced when outcomes are variable, rather than predictable	☐ Directive: People explicitly tell you they don't want a task automated or augmented
☐ {add your own}	☐ {add your own}

Data Preparation

- design new interfaces for lay users to interact, understand, and manipulate datasets and models



Data Preparation

Use data that applies to different groups of users

Your training data should reflect the diversity and cultural context of the people who will use it. Use tools like [Facets](#) and [WIT](#) to explore your dataset and better understand its biases. In doing so, note that to properly train your model, you might need to collect data from equal proportions of different user groups that might not exist in equal proportions in the real world. For example, to have speech recognition software work equally on all users in the United States, the training dataset might need to contain 50% of data from non-native English speakers even if they are a minority of the population.

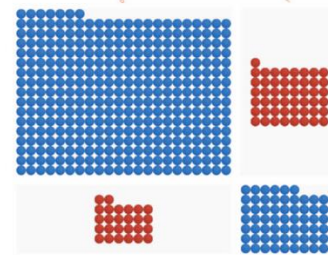
<https://pair-code.github.io/what-if-tool/>

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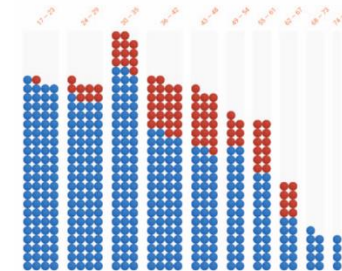
IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 26, NO. 1, JANUARY 2020

The What-If Tool: Interactive Probing of Machine Learning Models

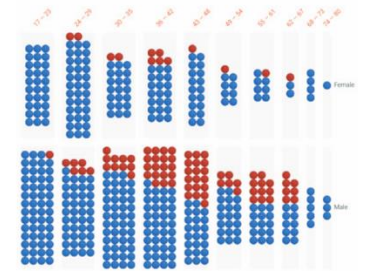
James Wexler, Mahima Pushkarna, Tolga Bolukbasi, Martin Wattenberg, Fernanda Viégas, and Jimbo Wilson



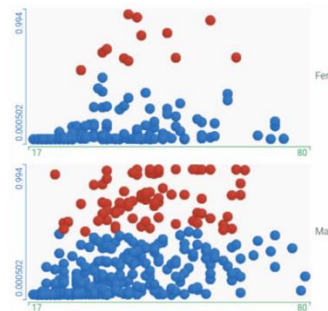
(a) Confusion matrix of a single binary classification model, colored by prediction correctness



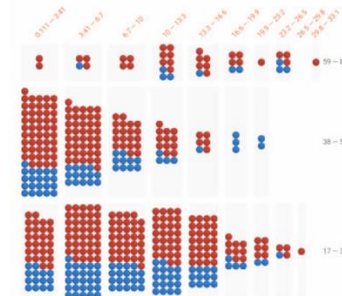
(b) Histogram of age, colored by classification



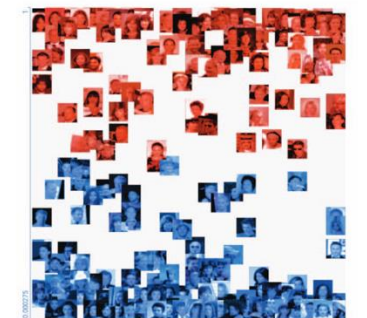
(c) Two-dimensional histogram of age and sex, colored by classification



(d) Small multiples by sex. Each scatterplot shows age vs positive classification score, colored by classification



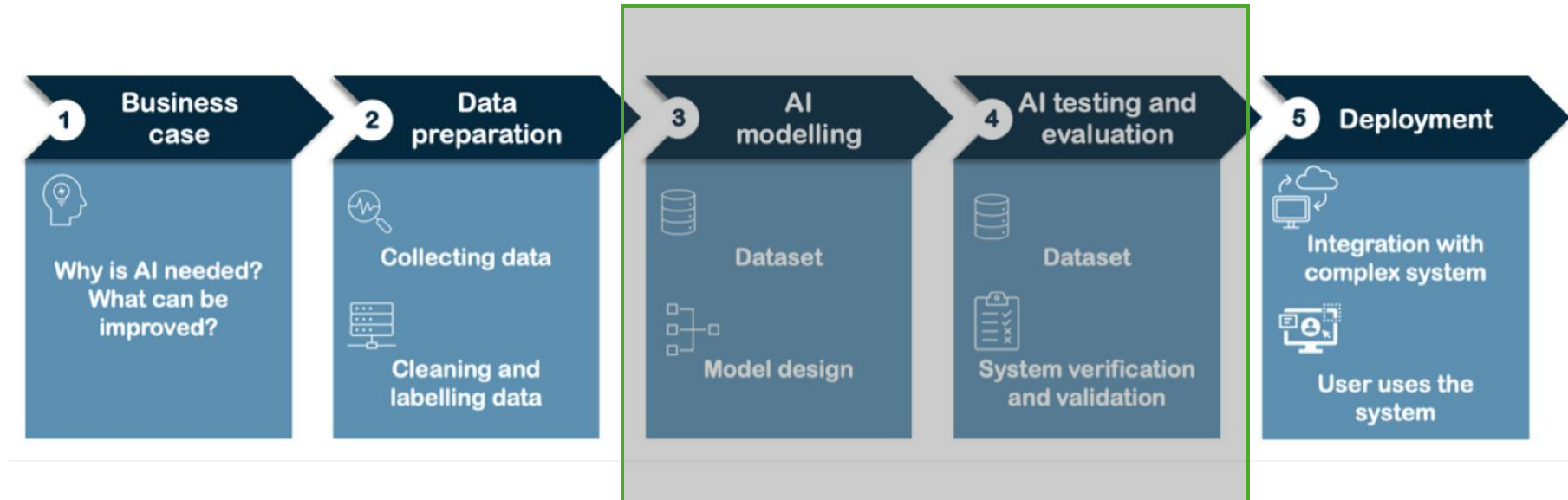
(e) Histograms of performance in a regression model that predicts age, faceted into 3 age buckets



(f) Using images as thumbnails for image datasets

Model building and evaluation

Allow users to shape the algorithms or e.g. develop interactive machine learning approaches that can take lay user feedback into account



Model building and evaluation

Include Explanations to participants + receive their feedback

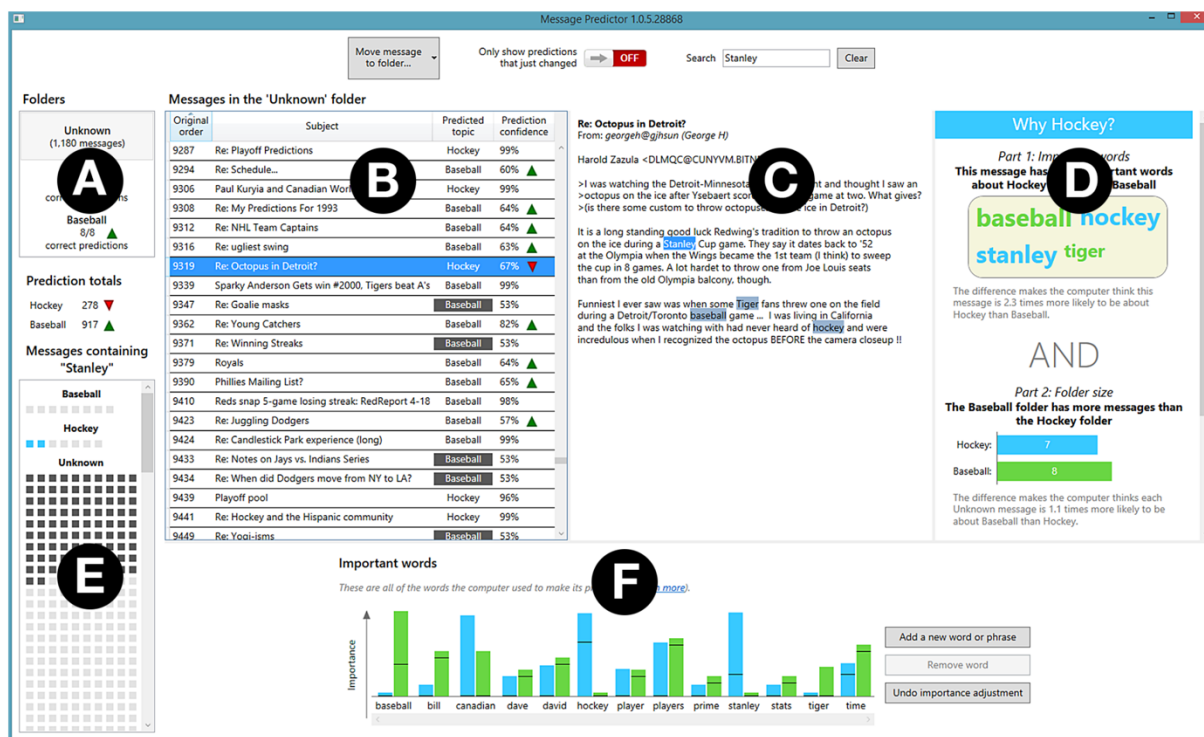


Figure 1. The Lucidebug prototype. (A) List of folders. (B) List of messages in the selected folder. (C) The selected message. (D) Explanation of the selected message's predicted folder. (E) Overview of which messages contain the selected word. (F) Complete list of words the learning system uses to make predictions.

Why Hockey?

Part 1: Important words

This message has more important words about Hockey than about Baseball

baseball hockey stanley tiger

The difference makes the computer think this message is 2.3 times more likely to be about Hockey than Baseball.

AND

Part 2: Folder size

The Baseball folder has more messages than the Hockey folder



The difference makes the computer thinks each Unknown message is 1.1 times more likely to be about Baseball than Hockey.

YIELDS

67% probability this message is about Hockey

Combining 'Important words' and 'Folder size' makes the computer think this message is 2.0 times more likely to be about Hockey than about Baseball.

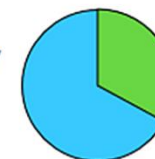


Figure 2. The Why explanation tells users how features and folder size were used to predict each message's topic. This figure is a close-up of Figure 1 part D.

Model building and evaluation

Co-design/imagine effects with users before deploying

Leverage participatory or speculative design methods

Consider using participatory approaches from [speculative](#) and [critical](#) design practices to evaluate potential downstream effects of your AI model. These approaches will provoke discussion about the kinds of outcomes you want to enable with your AI model, and help you identify risks and behaviors that you want to avoid or mitigate. Use these to prioritize short-term UX interventions while planning long-term strategies across your product's lifecycle.

Q&A: Participatory Machine Learning

<https://medium.com/people-ai-research/participatory-machine-learning-69b77f1e5e23>

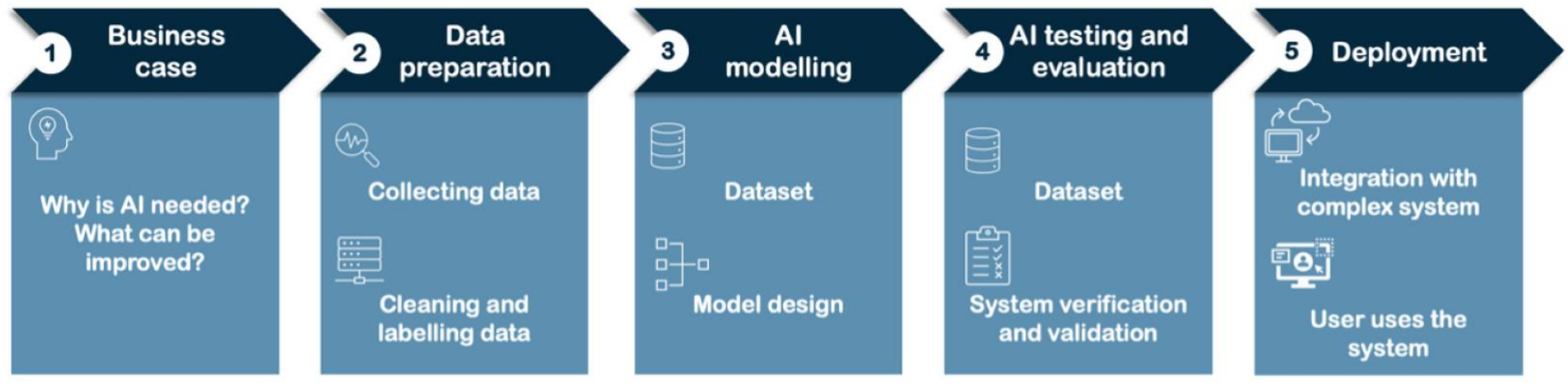
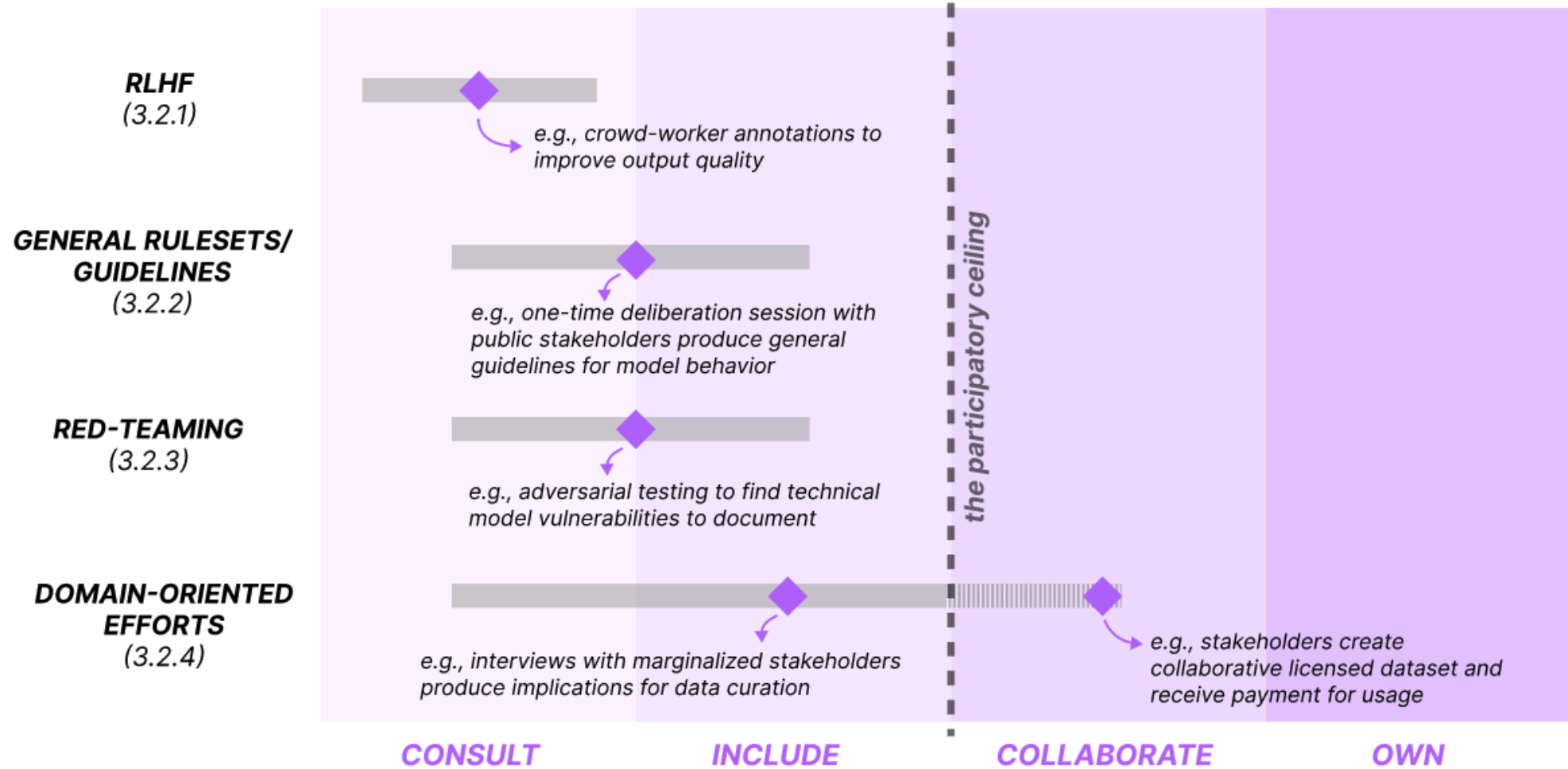


Fig. 8. AI system development process.

Participation in the Age of Foundation Models

- Reinforcement learning with human feedback (RLHF).
- Rulesets and policies
- Red teaming
- Domain-oriented efforts

Participation in the Age of Foundation Models



Human-centred AI as a Design Process

Part 1: Understanding user needs
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